# Customer Shopping Behavior Analysis Report

1. **Problem :**

A leading retail company aims to gain deeper insights into customer shopping behavior to enhance sales performance, improve customer satisfaction, and strengthen long-term customer loyalty. The company has observed shifting purchasing trends across different demographic groups, product categories, and shopping channels, including online and offline platforms. Management wants to identify the key factors influencing customer purchasing decisions and repeat buying behavior, such as discounts, product reviews, seasonal trends, and preferred payment methods.

2. **Tools Used :**

Python - Load dataset and perform Exploratory Data Analysis(EDA)

SQL -

Power Bi -Build an interactive dashboard that highlights key patterns and trends

3. **Dataset Summary:**

**Total Rows:** 3900

**Total Columns:** 18

- **Key Features:**

  **-Customer Demographics :-** Age, Gender, Location, Subscription Status

  **-Purchase Details :-** Item Purchased, Category, Purchase Amount, Season, Size, Color

  **-Shopping Behavior:-** Discount Applied, Promo code Used, Previous Purchases,Frequency of Purchases , Review Rating, Shipping Type

  **-Missing Data:-** 37 Values in Review Rating Column

**Frequency of purchasing:**

- How often a customer buys a product or service within a specific time period.

**Previous purchase** refers to the product or service a customer bought before the current transaction. Businesses use it to analyze buying patterns and improve recommendations

4. **Data Preparation and Modeling using Python(EDA):**

we began with data preparation and cleaning in python:

- **Data Loading:** Imported the dataset using Pandas.
- **Initial Exploration:** Use **df.info ()** to check the structure and **df.describe()** for summary statistics.

| | Customer ID | Age | Purchase Amount (USD) | Review Rating | Previous Purchases |
|---|---|---|---|---|---|
| **count** | 3900.000000 | 3900.000000 | 3900.000000 | 3863.000000 | 3900.000000 |
| **mean** | 1950.500000 | 44.068462 | 59.764359 | 3.750065 | 25.351538 |
| **std** | 1125.977353 | 15.207589 | 23.685392 | 0.716983 | 14.447125 |
| **min** | 1.000000 | 18.000000 | 20.000000 | 2.500000 | 1.000000 |
| **25%** | 975.750000 | 31.000000 | 39.000000 | 3.100000 | 13.000000 |
| **50%** | 1950.500000 | 44.000000 | 60.000000 | 3.800000 | 25.000000 |
| **75%** | 2925.250000 | 57.000000 | 81.000000 | 4.400000 | 38.000000 |
| **max** | 3900.000000 | 70.000000 | 100.000000 | 5.000000 | 50.000000 |

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

- **Missing Data Handling:** Checked for null values using **df.isnull().sum()** and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:**

  -Converts all **column names** to lowercase. This is a common step in data cleaning to ensure consistency and avoid issues caused by differing capitalization (e.g., 'Column Name' vs. 'column name').

  -Replaces any spaces within the column names with underscores. This practice makes column names more convenient to work with in Python.

  -I have replace space with hyphen instead of underscore now I have to replace hyphen with underscore because there is no space left anymore, spaces already changed with hypen(-).

  -Rename the column name using **rename function.**

- **Feature Engineering:**

  -Segmented customer **ages** into distinct **age_group** bins.

  -Created **purchase *frequency_days*** *column from purchase data.*

- **Data Consistency Check:**

  -Remove **promo_code_used** column because it has same information as in **discount** column. Reduce redundancy and storage.

## 5. Data Analysis using SQL:

We performed structured analysis in **Postgre SQL** to answer key business questions:

- **Data Connection:**

create database ----> Give name of the Database ----> Save---> Ingestion(Data successfully loaded into table 'customer' in database 'customer_behavior'.)

Data is there or not

Database ---> Schemas ---> Table ---> Customer Table

How do you know data in customer table or not ?

Database ---> Customer_behavior---> Query Tool--->Write simple query to check---> run

- **Revenue by Gender -**

| | gender text | sum numeric |
|---|---|---|
| 1 | Female | 75191 |
| 2 | Male | 157890 |

- **High Spending Discount users -**

| | customer_id bigint | purchase_amount bigint |
|---|---|---|
| 1 | 2 | 64 |
| 2 | 3 | 73 |
| 3 | 4 | 90 |
| 4 | 7 | 85 |
| 5 | 9 | 97 |
| 6 | 12 | 68 |
| 7 | 13 | 72 |
| 8 | 16 | 81 |
| 9 | 20 | 90 |
| 10 | 22 | 62 |
| 11 | 24 | 88 |
| 12 | 29 | 94 |
| 13 | 32 | 79 |
| 14 | 33 | 67 |
| 15 | 35 | 91 |
| 16 | 37 | 69 |
| 17 | 40 | 60 |
| 18 | 41 | 76 |
| 19 | 43 | 100 |
| 20 | 44 | 69 |
| 21 | 663 | 98 |
| 22 | 55 | 94 |
| 23 | 57 | 73 |

- **Top 5 Product by Rating -**

| | item_purchased text | Average Product Rating numeric |
|---|---|---|
| 1 | Gloves | 3.86 |
| 2 | Sandals | 3.84 |
| 3 | Boots | 3.82 |
| 4 | Hat | 3.80 |
| 5 | Skirt | 3.78 |

- **Shipping type comparison -**

| | shipping_type text | Avg Purchase Amount numeric |
|---|---|---|
| 1 | Standard | 58.46 |
| 2 | Express | 60.48 |

- **Subscribers vs. Non subscribers -**

| | subscription_status text | count bigint | avg_spend numeric | total_revenue numeric |
|---|---|---|---|---|
| 1 | Yes | 1053 | 59.49 | 62645 |
| 2 | No | 2847 | 59.87 | 170436 |

- **Discount dependent product -**

| | item_purchased text | discount_rate numeric |
|---|---|---|
| 1 | Hat | 50.00 |
| 2 | Sneakers | 49.00 |
| 3 | Coat | 49.00 |
| 4 | Sweater | 48.00 |
| 5 | Pants | 47.00 |

- **Customer Segmentation -**

| | customer_segment text | number_of_customers bigint |
|---|---|---|
| 1 | Loyal | 3116 |
| 2 | New | 83 |
| 3 | Returning | 701 |

- **Top 3 Product by Category -**

| | item_rank bigint | category text | item_purchased text | total_orders bigint |
|---|---|---|---|---|
| 1 | 1 | Accessori... | Jewelry | 171 |
| 2 | 2 | Accessori... | Sunglasses | 161 |
| 3 | 3 | Accessori... | Belt | 161 |
| 4 | 1 | Clothing | Blouse | 171 |
| 5 | 2 | Clothing | Pants | 171 |
| 6 | 3 | Clothing | Shirt | 169 |
| 7 | 1 | Footwear | Sandals | 160 |
| 8 | 2 | Footwear | Shoes | 150 |
| 9 | 3 | Footwear | Sneakers | 145 |
| 10 | 1 | Outerwear | Jacket | 163 |
| 11 | 2 | Outerwear | Coat | 161 |

- **Repeat Buyers and Subscriptions -**

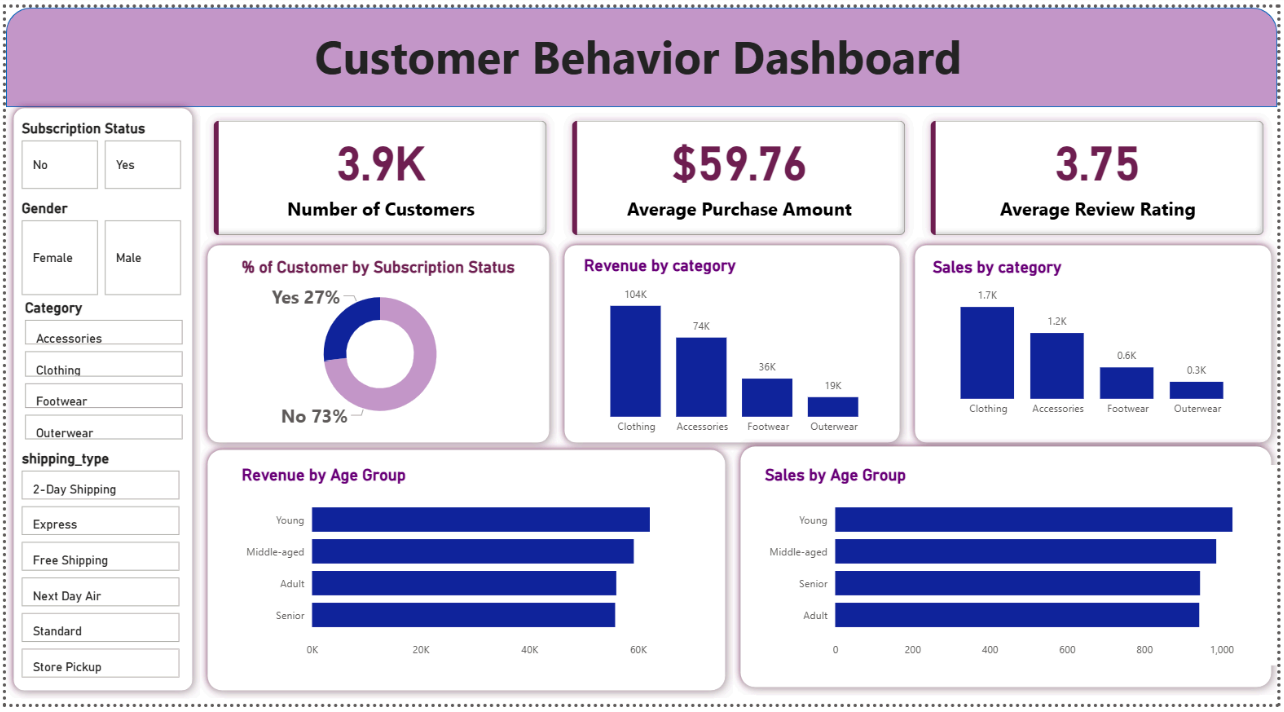
| | subscription_status text | repeated_buyers bigint |
|---|---|---|
| 1 | No | 2518 |
| 2 | Yes | 958 |

- **Revenue by Age Group -**

| | age_group text | total_revenue numeric |
|---|---|---|
| 1 | Young | 62143 |
| 2 | Middle-aged | 59197 |
| 3 | Adult | 55978 |
| 4 | Senior | 55763 |

6. **Dashboard in Power BI**

Finally, we built an interactive dashboard in Power BI to present insights visually.




7. **Business Recommendations**

**-Boost Subscriptions** - Implement a premium subscription tier featuring high-value, exclusive rewards.

**-Customer Loyality Program -** Design and execute an exclusive loyalty ecosystem to reward high-value shoppers.

**-Product Positioning -** Highlight top-rated and high selling products in campaigns.

**-Targeted Marketing -** Target high age group people and express shipping user.